

Large Language Models are Few-Shot Health Learners

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1. Contents

1. What is LLM?

대형 언어 모델(Large Language Model, LLM)

- 대규모 데이터셋으로 학습되었으며, 대규모 파라미터를 가진 모델
- 기존의 PLM(사전학습모델)들을 대규모로 확장하면 성능이 오른다는 것을 확인 -> ‘대형’의 기준은 없음
- (GPT-3의 학습 파라미터 개수 1750억 개)
- 방대한 양의 텍스트 데이터를 사전 학습(Pre-trained)한 전이 학습(Transfer) 모델임.
- ChatGPT, LLaMA : 생성형 언어 모델, Decoder 기반 언어 모델

Basic Term

Prompt engineering : 모델에 입력하는 질문이나 명령을 구조화하여 모델의 성능을 향상시키는 과정

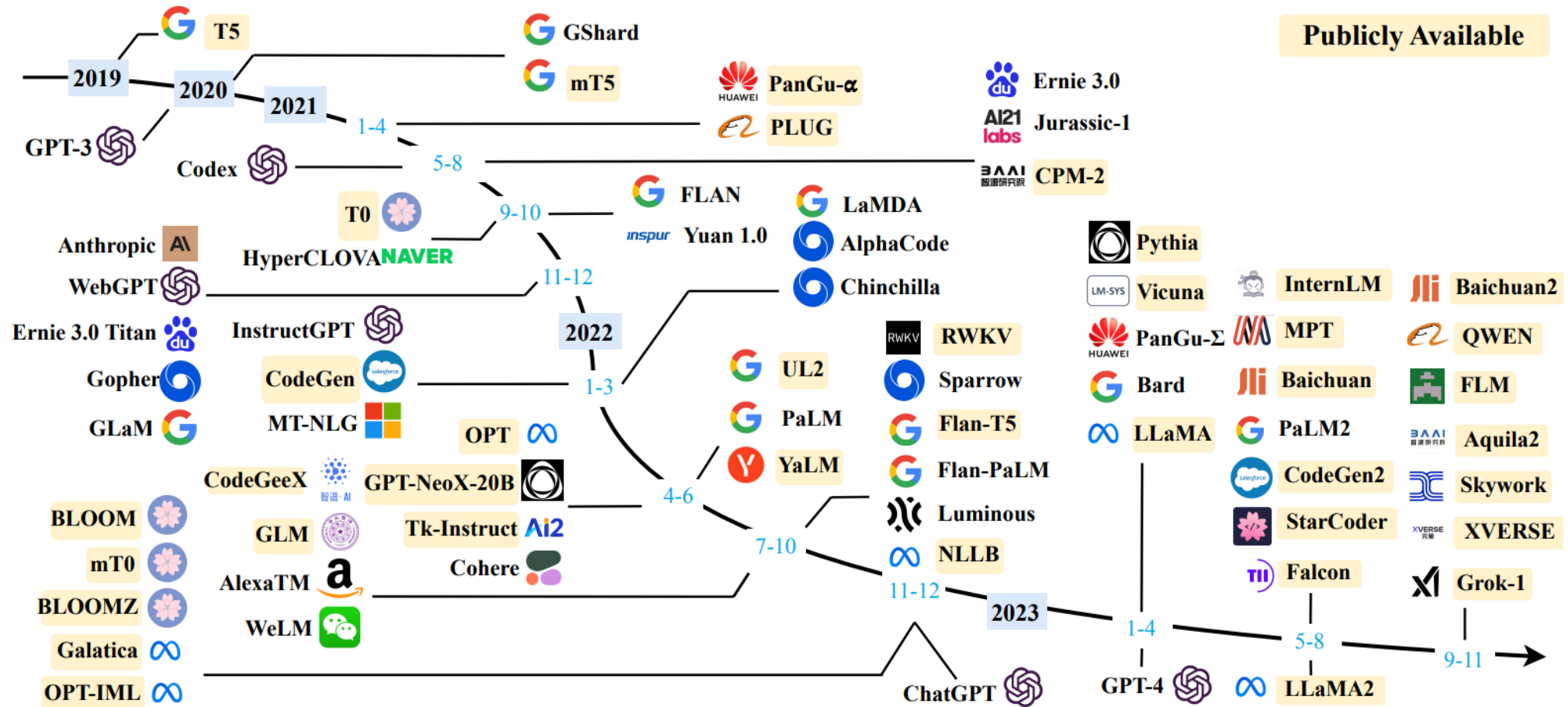
Prompt tuning

Foundation model : Label이 지정되지 않는 광범위한 데이터 집합에 대해 훈련된 대규모 인공 지능 모델, 광범위한 다운스트림 작업에 적용할 수 있는 AI 모델 유형을 말함

1. Contents

1. What is LLM?

대형 언어 모델(Large Language Model, LLM)



1. Introduction

Large Language Models are Few-Shot Health Learners

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Abstract

Large language models (LLMs) can capture rich representations of concepts that are useful for real-world tasks. However, language alone is limited. While existing LLMs excel at text-based inferences, health applications require that models be grounded in numerical data (e.g., vital signs, laboratory values in clinical domains; steps, movement in the wellness domain) that is not easily or readily expressed as text in existing training corpus. We demonstrate that with only few-shot tuning, a large language model is capable of grounding various physiological and behavioral time-series data and making meaningful inferences on numerous health tasks for both clinical and wellness contexts. Using data from wearable and medical sensor recordings, we evaluate these capabilities on the tasks of cardiac signal analysis, physical activity recognition, metabolic calculation (e.g., calories burned), and estimation of stress reports and mental health screeners.

1 Introduction

The fundamental ability of large language models (LLMs) to capture knowledge and concepts has improved dramatically as neural architectures have been scaled. Exponentially greater numbers of parameters and training samples has led to models that can encode vast amounts of information [3, 5, 19, 24, 25]. These models present opportunities in many domains including software engineering [4], information retrieval and data mining [10], content creation [6] and health [22]. However, in many cases text alone is not enough. Grounding in non-linguistic observations such as actions, experiences or perceptions is an essential part of deriving meaning from language [8] as words alone does not capture all the necessary information to complete a given task. Tuning, or grounding, of this kind is necessary to connect the knowledge contained within a language model to events or actions in the physical world [1]. For these reasons, models have been combined with data from other modalities, most commonly images [11, 16, 18], audio [27], video [23, 30] or physical actions [1].

Despite the importance of physiology and behavior to human health, evaluating and grounding models with these data has been relatively unexplored. Medical-domain language models have shown remarkable capabilities to capture sophisticated knowledge [10, 22], performing well on multiple-choice examination questions (e.g., board exams). However, they have not been tested extensively on tasks that involve analysis of physiological (e.g., heart rate) and behavioral timeseries (e.g., daily steps), which are important for many consumer health tasks. Without data of this kind, derived from actual measurements of the human body, these models may be limited in their ability to provide specific health insights, particularly when the insight involves inference from physiological data. For example, an individual searching for a diagnosis for their ailment may have limited success if the model has no insight into physiology (e.g., elevated heart rate, fever) that could be underlying their symptoms. Conversely, models capable of incorporating user physiological data may be able to provide enhanced insight to queries within a medical domain. We hypothesize that LLMs have some capability to interpret and reason about health information, but that that ability will dramatically improve with grounding from even a small number of examples of numerical health data.

**“Large Language Models are Few-Shot Health Learners”,
Consumer Health Research Team, Google**

Xin Liu, Daniel McDuff, et al

Citation : 44, arXiv

1. Introduction

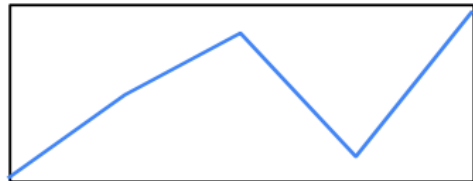
1. Question-answer pairs for health tasks

Activity Recognition

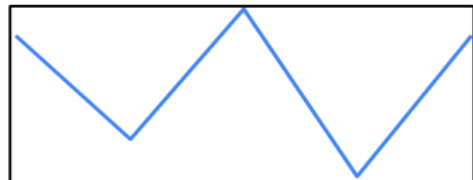
Prompt:

Classify the accelerometer data in meters per second squared as either walking or running.

Response:



Walking.



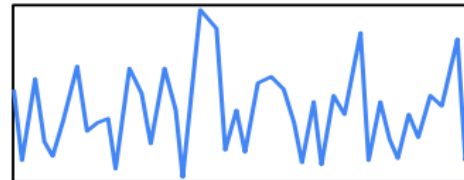
Running.

Atrial Fibrillation Classification

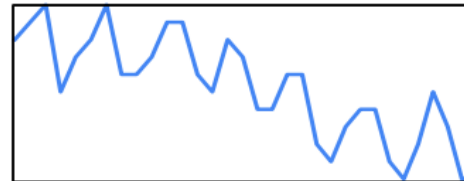
Prompt:

Classify the given Interbeat Interval sequence in ms as either Atrial Fibrillation or Normal Sinus.

Response:



Atrial Fibrillation.



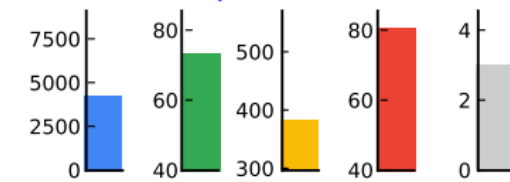
Normal Sinus.

Stress

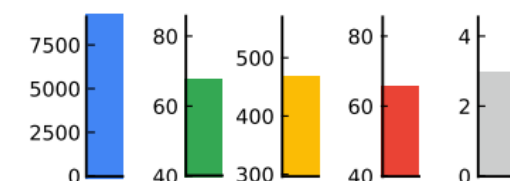
Prompt:

Steps [Steps], resting heart rate: [RHR] beats/min, sleep duration: [SleepMinutes] mins, non-rem heart rate: [NREMHR] beats/min, mood last day [Mood] out of 5. What will my stress level be?

Response:



Stress: 5 out of 5.



Stress: 1 out of 5.

■ Steps ■ NREM HR ■ Sleep Mins ■ Rest. HR ■ Mood Prev. Day

1. Introduction

1. Abstract

Novelty

1. Curate a consumer health tasks
2. Demonstrate the capability of LLM
3. Expose the limitations of LLM
4. Present an set of tuned models

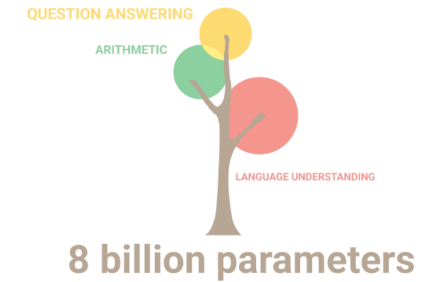
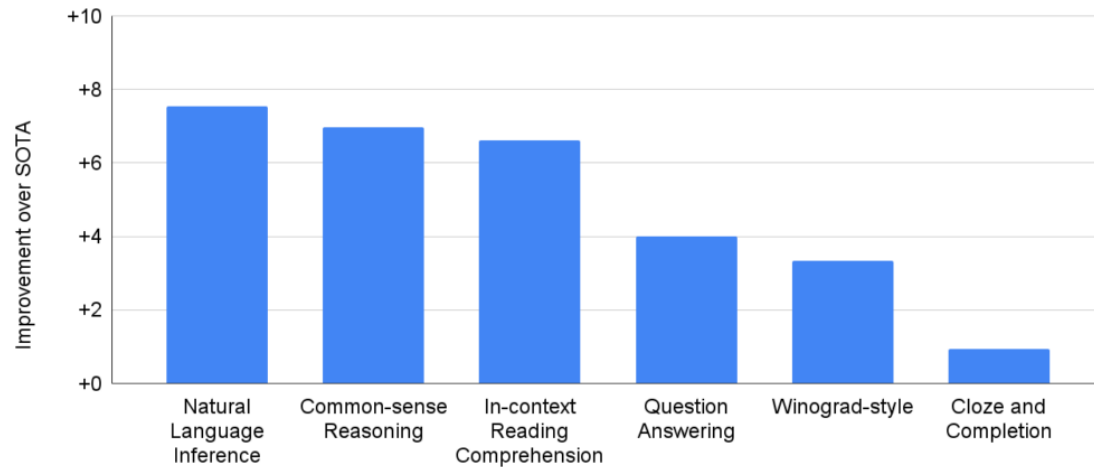
Topic	Data	Task	Prompt	Target
Cardiovascular	MIT-BIH & MIMIC-III [9, 12]	Instantaneous HR to Average HR	Given a sequence of heart rates, calculate the average heart rate: [HR]	$\mu([HR])$
		IBIs to HR	Calculate heart rate based on the given inter-beat intervals in ms: [IBI]	$\frac{60}{\mu([IBI])}$
		IBIs to Atrial Fibrillation	Classify the given Interbeat Interval sequence in ms as either Atrial Fibrillation or Normal Sinus: [IBI]	[A.Fib.]
		IBIs to Sinus Bradycardia	Classify the given Interbeat Interval sequence in ms as either Sinus Bradycardia or Normal Sinus: [IBI]	[Sinus. B.]
		IBIs to Sinus Tachycardia	Classify the given Interbeat Interval sequence in ms as either Sinus Tachycardia or Normal Sinus: [IBI]	[Sinus T.]
Activity	PAMAP2 [21]	Activity Recognition	Classify the following accelerometer data in meters per second squared as either walking or running: [Acc]	[Activity]
Metabolic	Synthetic	Calorie Estimate	How many total calories will I burn after [Activity] for [Duration] minutes? My weight is [Weight] lbs.	[Calories]
MHealth	Fitbit	Estimation of Daily Stress EMA	Steps: [Steps], resting heart rate: [RHR] beats/min, sleep duration: [SleepMinutes] minutes, non-REM heart rate [NREMHR] beats/min, mood last day [Mood] out of 5. What will my stress level be?	[Stress]
		Estimation of PHQ8 Score	Steps: [Steps], resting heart rate: [RHR] beats/min, sleep duration: [SleepMinutes] minutes, non-REM heart rate [NREMHR] beats/min. feeling last month: [Mood] out of 5. What will my PHQ score be?	[PHQ]

IBI = Interbeat Interval, HR = Heart Rate, RHR = Resting HR, NREMHR = Non-REM HR, PHQ = Patient Health Questionnaire, EMA = Ecological Momentary Assessment

2. Method

1. Architecture

PaLM : Google pretrained model(54 billion parameter – transformer architecture)

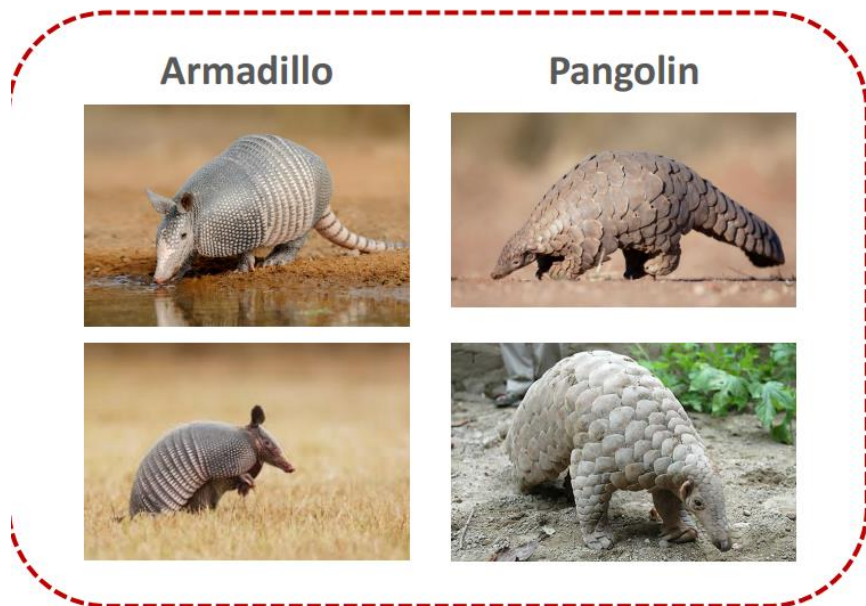


2. Method

1. What is few-shot learning?

“Few한 데이터도 잘 분류할 수 있다는 것” -> 즉 구분하는 방법을 배우는 것

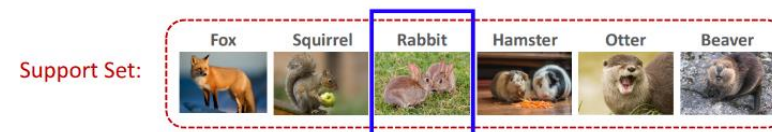
Support Set



Query



Armadillo or Pangolin?



Training set에 없는 클래스를 맞추는 문제.

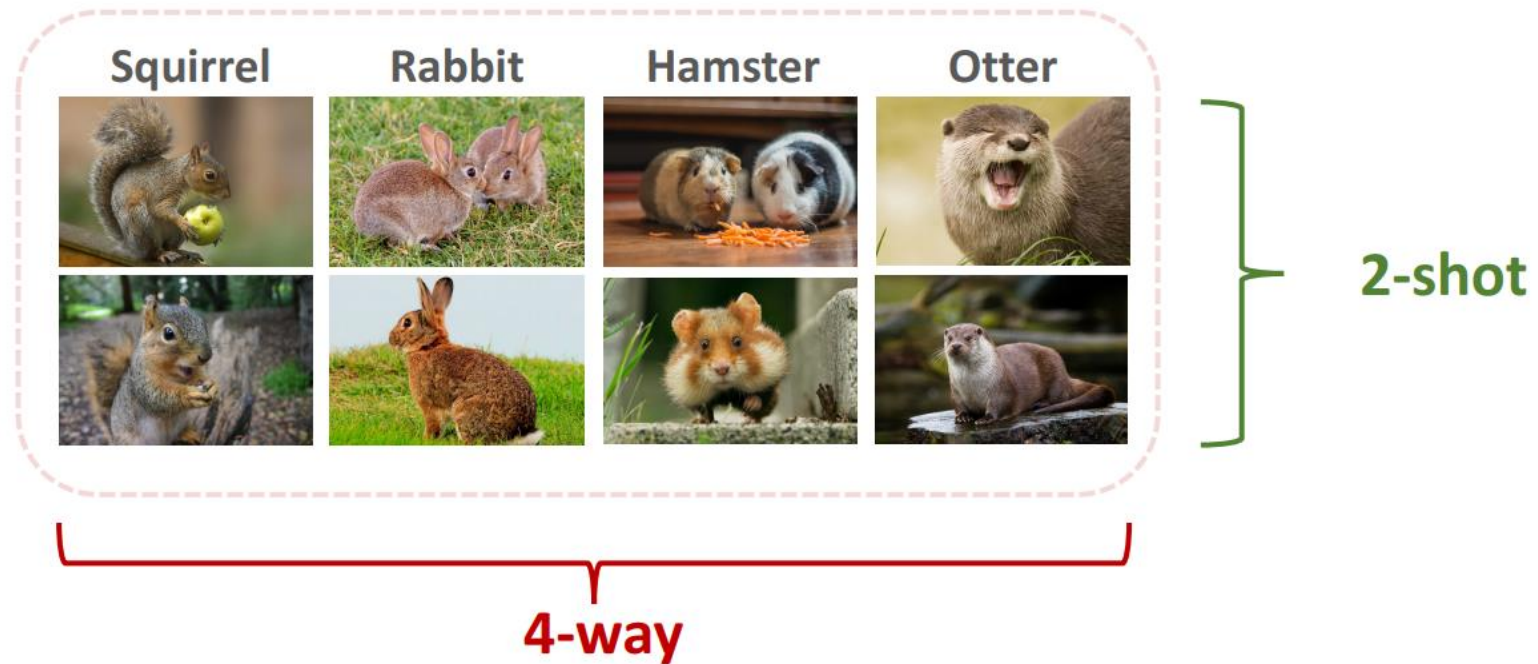
2. Method

1. What is few-shot learning?

K-way n-shot

K : class, n : sample size

Support Set:



2. Method

Dataset

Cardiovascular

- HRs to Average HR
- IBIs(interbeat intervals) to HR
- IBIs(interbeat intervals) to A.Fib(atrial fibrillation).
- IBIs(interbeat intervals) to Sinus B.(sinus bradycardia)
- IBIs(interbeat intervals) to Sinus T(sinus tachycardia).

Activity

- IMU(Inertial Measurement Unit) Activity – walking, running

Metabolic

- Calories $Calories = MET\ Value * Duration * Weight \div 200$

Mhealth

- Fitbit to Stress
- Fitbit to PHQ(Patient health questionnaire)

Fitbit metrics (step count, resting heart rate and heart rate during non-REM sleep and sleep duration)

2. Method

1. Zero-Shot learning

Input : “Classify the following accelerometer data in meters per second squared as either walking or running:
0.052, 0.052, 0.052, 0.051, 0.052, 0.055, 0.051, 0.056, 0.06, 0.064”

Label : “Running”

Chatgpt4o

Input : Same textual templates

Label : “walking”

“Classify the following accelerometer data in meters per second squared as either walking or running: 0.052, 0.052, 0.052, 0.051, 0.052, 0.055, 0.051, 0.056, 0.06, 0.064”

1. Average Value:

- Calculate the mean (average) of the data points.

$$\text{Mean} = \frac{0.052 + 0.052 + 0.052 + 0.051 + 0.052 + 0.055 + 0.051 + 0.056 + 0.060 + 0.064}{10}$$

2. Range and Variability:

- Find the range (difference between the maximum and minimum values).

$$\text{Range} = 0.064 - 0.051 = 0.013 \text{ m/s}^2$$

3. Standard Deviation:

- Calculate the standard deviation to understand the variability.

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

Where μ is the mean value, and n is the number of data points.

$$\text{Variance} = \frac{(0.052 - 0.0545)^2 + (0.052 - 0.0545)^2 + (0.052 - 0.0545)^2 + (0.051 - 0.0545)^2 + (0.052 - 0.0545)^2 + (0.055 - 0.0545)^2 + (0.051 - 0.0545)^2 + (0.056 - 0.0545)^2 + (0.060 - 0.0545)^2 + (0.064 - 0.0545)^2}{10}$$
$$\text{Variance} \approx 0.00001575 \text{ (simplified)}$$
$$\text{Standard Deviation} = \sqrt{\text{Variance}} \approx \sqrt{0.00001575} \approx 0.004 \text{ m/s}^2$$

4. Interpretation:

- Walking typically has lower acceleration values (around 0.5 to 1.5 m/s²) with less variability.
- Running has higher acceleration values (around 2.0 to 3.5 m/s²) with more variability.

However, if we must choose between walking and running based on the given data:

- The low average acceleration value and low variability (standard deviation) suggest that the activity is not vigorous. This indicates a higher likelihood of walking over running.

Conclusion:

Based on the provided accelerometer data, it is classified as **walking**.

2. Method

Training Methods

1. Zero-Shot
2. Prompt Eng
3. Prompt Tune

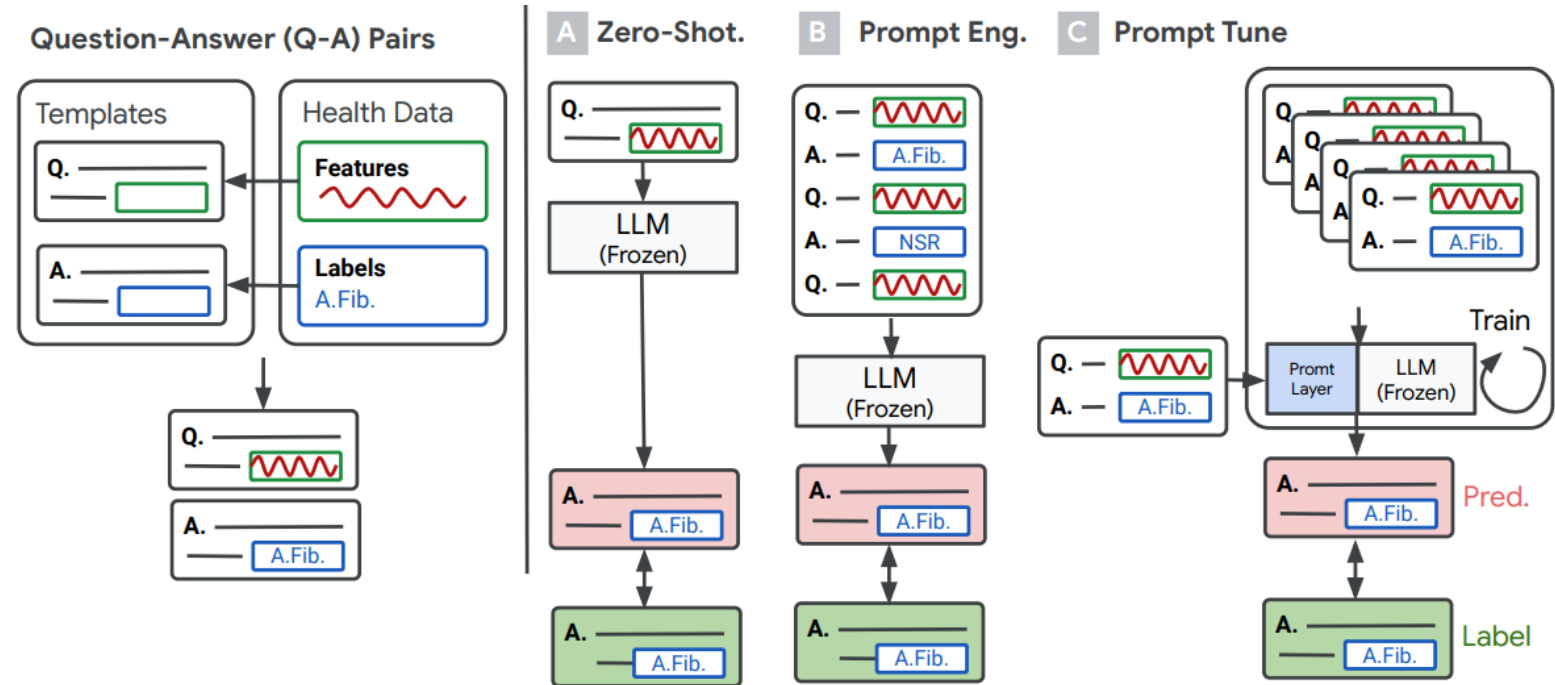
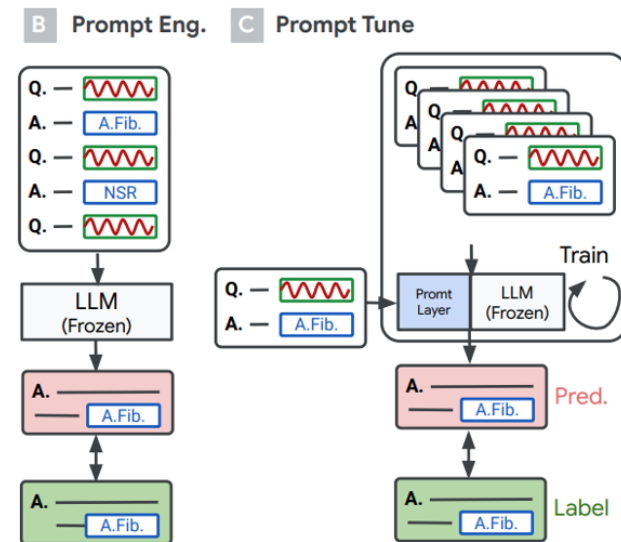


Figure 2: **Our dataset construction and training configurations.** We use physiological data to construct question-answer pairs. We create training, validation and test splits and compare zero-shot prediction performance to models created via prompt engineering and prompt tuning.

2. Method

Prompt Engineering vs Prompt tuning

	Prompt Tun	Prompt Eng
개념	프롬프트를 벡터화하여 자동으로 최적화 하는 기법	모델에게 주어지는 입력 문장을 수작업으로 설계하는 기법
접근 방식	머신러닝, 임베딩 기법 사용	경험과 직관을 기반으로 수작업으로 설계
프롬프트 표현	연속형 벡터 임베딩	텍스트 형태의 입력
효율성	모델 재훈련 없이 성능 개선	모델 재훈련 없이 성능 개선
도메인 지식 활용	도메인 지식 필요 x	도메인 지식 필요
적용 사례	복잡한 작업 성능 향상	초기 단계의 간단한 작업 수행 혹은 특정 도메인 작업
유연성	특정 작업에 최적화된 경우 유용	다양한 작업에 유연하게 적용 가능



2. Method

2. Prompt tuning

- 1) The Numerical-only
- 2) The Context-Inclusive

Table 2: **Results.** Comparison of performance between prompt-tuned LLMs (w/ Context-Inclusive Prompts) and supervised neural network training across all consumer health tasks.

Topic	Task	Metric	Supervised Baseline			LLM with Context			% Improvement
			3-Shot	10-Shot	25-Shot	3-Shot	10-Shot	25-Shot	
Cardio	HRs to Average HR	MAE ↓ (beats/min)	3.41	1.37	1.08	6.00	2.49	1.06	+1.90%
	IBIs to HR	MAE ↓ (beats/min)	34.0	20.0	19.8	12.3	5.87	5.01	+74.7%
	IBIs to A.Fib.	Accuracy ↑ (%)	52.5	72.5	75.0	85.0	75.0	89.0	+19.7%
	IBIs to Sinus B.	Accuracy ↑ (%)	88.0	86.0	86.0	81.0	79.0	92.0	+7.00%
	IBIs to Sinus T.	Accuracy ↑ (%)	56.0	53.0	61.0	65.0	82.0	88.0	+44.3%
Activity	IMU Activity	Accuracy ↑ (%)	56.0	60.0	64.0	62.0	80.0	85.0	+32.8%
Metabolic	Calories	MAE ↓ (calories)	185	97	89	106	77	48	+46.1%
MHealth	Fitbit to Stress	Accuracy ↑ (%)	37.5	70.5	80.0	72.5	71.5	82.5	+3.10%
	Fitbit to PHQ	Accuracy ↑ (%)	51.0	52.0	53.0	49.0	59.0	69.0	+30.2%

2. Method

3. Prompt Engineering

- Incorporate additional context

Q : How many total calories will I burn after walking for 50 minutes? My weight is 156 lbs.

A : 214

Q : How many total calories I burned after walking for 48 minutes? My weight is 221 lbs.

A : “ ”

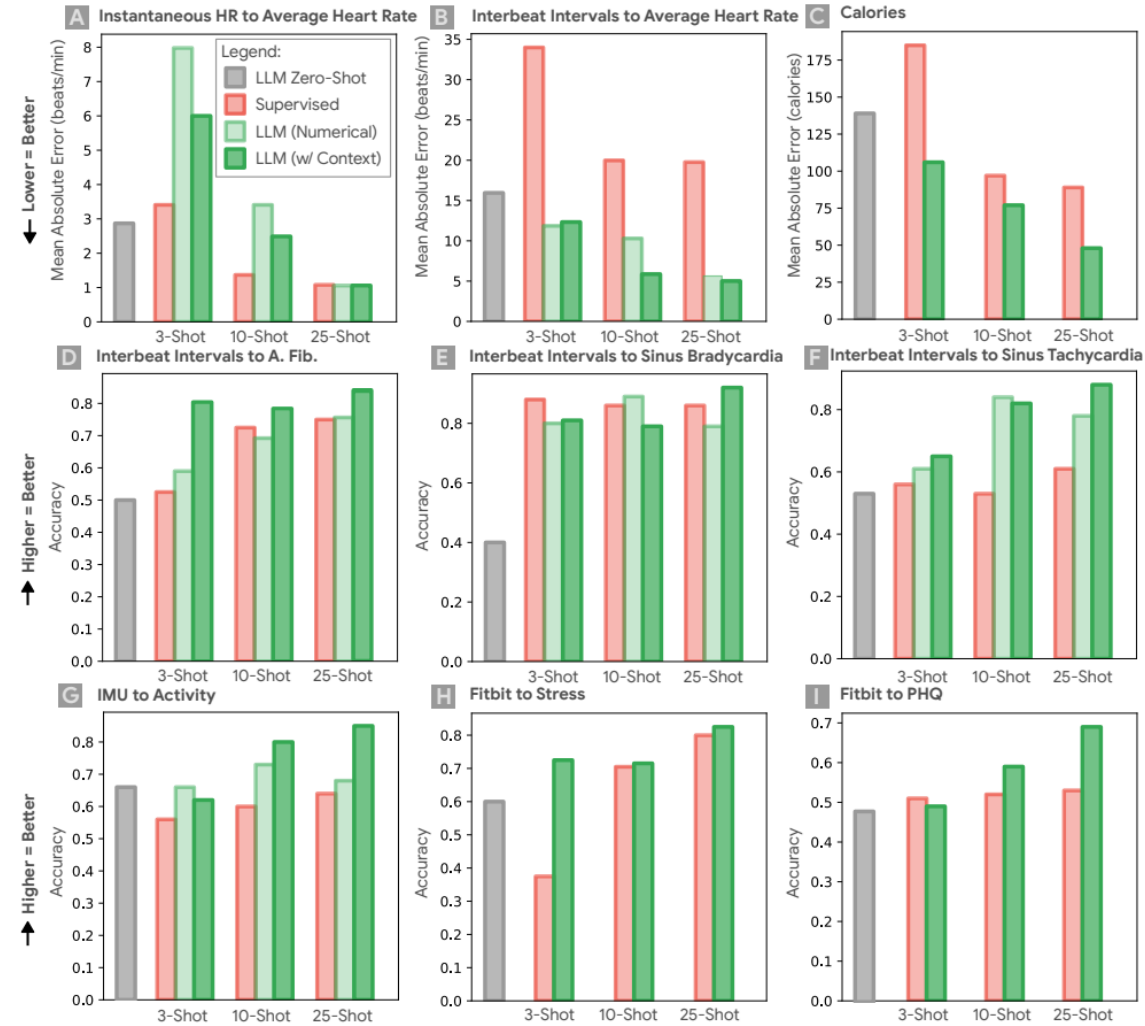
4. Supervised Baseline

Multilayer Perceptron(MLP)

Layer size : 4096 * 1

Setting : 32 TPU-v4

3. Results



3. Results

Does offering more context to the LLM enhance performance?

Table 3: **Numerical-Only vs Context-Inclusive.** Performance on tasks when the input prompt contains solely numerical values (Numerical-Only Prompts) versus when it includes textual context (Context-Inclusive Prompts).

Topic	Task	Metric	Performance		
			No Context	Context	% Improvement
Cardio	HRs to Average HR	MAE ↓ (beats/min)	1.06	1.06	0%
	IBIs to HR	MAE ↓ (beats/min)	5.63	5.01	+11.0%
	IBIs to A.Fib.	Accuracy ↑ (%)	75.6	89.0	+17.7%
	IBIs to Sinus B.	Accuracy ↑ (%)	79.0	92.0	+16.5%
	IBIs to Sinus T.	Accuracy ↑ (%)	78.0	88.0	+12.8%
Activity	IMU to Activity	Accuracy ↑ (%)	68.0	85.0	+25.0%

4. Conclusion

Conclusion

- ✓ 수치형 데이터에서 LLM의 모델의 한계를 보여줌
- ✓ 다른 수치 모델과의 비교는 부족함
- ✓ Few shot learning의 장점을 보여줌 with 생리학적, 행동학적 수치 데이터

THANK YOU
